KINGSCLIFF HIGH SCHOOL

ACHIEVEMENT THROUGH ENDEAVOUR

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MATHEMATICS

Year 9 5.2

Yearly Exam

2026

Marking Guide

Notes for markers

1. Full marks for a correct answer unless the question asks for working.
2. Follow-through: a later part that uses an earlier wrong answer correctly earns its marks.
3. Units: do not deduct for missing units; note it for feedback.
4. Rounding: deduct only where the criteria say so.
5. Green = Foundation, Blue = Standard (Test yourself levels).

Sections and Marks

Section 1 - Data
11 green · 6 blue /17

Section 2 - Surface area and volume
12 green · 10 blue /22

Section 3 - Mixed Topics
26 green · 7 blue /33

Total marks /72

Section	Marks	Share	Green	Blue	Questions
Section 1 - Data	17	23.6%	11	6	4 MC · 1 SA · 2 WP
Section 2 - Surface area and volume	22	30.6%	12	10	4 MC · 3 SA · 2 WP
Section 3 - Mixed Topics	33	45.8%	26	7	5 MC · 7 SA · 3 WP
Total	**72**	**100%**	**49 (68.1%)**	**23 (31.9%)**	

Section 1 answer summary – Data				
Q	**Answer**	**Marks**	**Level**	**Topic and source**
1	D (The time taken to swim 50 m)	1	Green	Classifying data TY9 Q9 (9.05, green)
2	B (Negatively skewed, clustered at 8 and 9)	1	Green	Shape of a distribution (dot plot) TY9 Q6a (9.03, green)
3	A (2)	1	Green	Mode from a histogram TY9 Q4b (9.02, green)
4	C (The mean)	1	Green	Effect of an outlier TY9 Q2c and Q3c-d (9.01, 9.02, green)
5a	25 minutes	2	Green	Mean and median from a stem-and-leaf plot TY9 Q3b (9.02, green)
5b	24 minutes	1	Green	Mean and median from a stem-and-leaf plot TY9 Q3b (9.02, green)
6a	16 students	1	Green	Reading a column graph TY9 Q4 (9.02, green)
6b	60 students	1	Green	Reading a column graph TY9 Q4 (9.02, green)
6c	40%	2	Green	Reading a column graph TY9 Q4 (9.02, green)
7a	Kestrels: 39; Tigers: 42.5	2	Blue	Comparing two data sets (back-to-back stem-and-leaf plot) TY9 Q7 (9.04, blue)
7b	The Tigers: their median (42.5) is higher than the Kestrels' (39).	1	Blue	Comparing two data sets (back-to-back stem-and-leaf plot) TY9 Q7 (9.04, blue)
7c	The Kestrels: their range is 15 goals (46 – 31), much smaller than the Tigers' 35 goals (57 – 22).	3	Blue	Comparing two data sets (back-to-back stem-and-leaf plot) TY9 Q7 (9.04, blue)
	Total	**17**	11 green · 6 blue	

Section 2 answer summary – Surface area and volume				
Q	**Answer**	**Marks**	**Level**	**Topic and source**
1	C (4600 L)	1	Green	Metric units: capacity (kL to L) TY10 Q1i (10.01, green)
2	A (±25 mL)	1	Green	Precision and absolute error TY10 Q2 (10.02, green)
3	D (72 cm^2)	1	Green	Area of a parallelogram TY10 Q6a (10.05, green)
4	B (176 cm^2)	1	Green	Area of a kite TY10 Q6b (10.05, green)
5a	36 cm	1	Green	Perimeter and area of a triangle TY10 Q4 and Q5 (10.04, green)

Section 2 answer summary – Surface area and volume (continued)				
Q	**Answer**	**Marks**	**Level**	**Topic and source**
5b	54 cm^2	1	Green	Perimeter and area of a triangle TY10 Q4 and Q5 (10.04, green)
6a	40.8 cm	1	Blue	Circumference and area of a circle TY10 Q7 (10.06, blue)
6b	132.7 cm^2	1	Blue	Circumference and area of a circle TY10 Q7 (10.06, blue)
7a	1696.5 cm^3	2	Blue	Volume and surface area of a cylinder TY10 Q12 and Q11 (10.09, 10.08, blue)
7b	791.7 cm^2	2	Blue	Volume and surface area of a cylinder TY10 Q12 and Q11 (10.09, 10.08, blue)
8a	60 m	1	Green	Perimeter and area of a composite shape (lawn) TY10 Q4a and Q5 (10.04, green)
8b	166 m^2	3	Green	Perimeter and area of a composite shape (lawn) TY10 Q4a and Q5 (10.04, green)
8c	$2058.40	1	Green	Perimeter and area of a composite shape (lawn) TY10 Q4a and Q5 (10.04, green)
9a	72 000 cm^3	1	Blue	Volume, capacity and surface area of a closed box TY10 Q14 (10.09, blue), Q1i (10.01, green) and Q9a (10.07, blue)
9b	72 L	1	Green	Volume, capacity and surface area of a closed box TY10 Q14 (10.09, blue), Q1i (10.01, green) and Q9a (10.07, blue)
9c	10 800 cm^2	3	Blue	Volume, capacity and surface area of a closed box TY10 Q14 (10.09, blue), Q1i (10.01, green) and Q9a (10.07, blue)
	Total	**22**	12 green · 10 blue	

Section 3 answer summary – Mixed Topics (Chapters 1–8)				
Q	**Answer**	**Marks**	**Level**	**Topic and source**
1	C (47)	1	Green	Substitution TY1 Q2 (1.02, green)
2	B (10.9)	1	Green	Pythagoras' theorem: hypotenuse TY2 Q5 (2.03, green)
3	C ($123.25)	1	Green	Discount and selling price TY3 Q16 (3.06, green)
4	A ($12m^8$)	1	Green	Index laws: multiplying terms TY5 Q1 (5.01, green)

Section 3 answer summary – Mixed Topics (Chapters 1–8) (continued)				
Q	**Answer**	**Marks**	**Level**	**Topic and source**
5	B ($3140.00)	1	Green	Salary: pay per fortnight TY8 Q1 (8.01, green)
6a	$3a + 4b$	1	Green	Simplifying algebraic expressions TY1 Q3 and Q4 (1.03, 1.04, green)
6b	$12xy$	1	Green	Simplifying algebraic expressions TY1 Q3 and Q4 (1.03, 1.04, green)
6c	$3n$	1	Green	Simplifying algebraic expressions TY1 Q3 and Q4 (1.03, 1.04, green)
7	$10x - 3$	2	Blue	Expanding and simplifying TY1 Q7 (1.07, blue)
8a	$3(2x + 5)$	1	Blue	Factorising TY1 Q8 (1.08, blue)
8b	$y(y - 8)$	1	Blue	Factorising TY1 Q8 (1.08, blue)
9	$913.50	2	Green	Simple interest TY3 Q19 (3.07, green)
10a	$132	1	Green	Profit as a percentage of the cost price TY3 Q17 (3.06, green)
10b	27.5%	1	Green	Profit as a percentage of the cost price TY3 Q17 (3.06, green)
11a	$x = 7$	2	Green	Solving equations TY7 Q1 (7.01, green) and Q5 (7.03, blue)
11b	$m = 8$	3	Blue	Solving equations TY7 Q1 (7.01, green) and Q5 (7.03, blue)
12	$1690	2	Green	Commission TY8 Q5 (8.03, green)
13a	$1018.40	1	Green	Wages and overtime TY8 Q2 and Q3 (8.01, 8.02, green)
13b	$1259.60	2	Green	Wages and overtime TY8 Q2 and Q3 (8.01, 8.02, green)
14a	$240	1	Green	Using a formula; solving an equation TY7 Q11 (7.06, green)
14b	16 weeks	2	Green	Using a formula; solving an equation TY7 Q11 (7.06, green)
15a	4.95 m	2	Green	Ladder: Pythagoras and trigonometry TY2 Q8 (2.04, green) and TY4 Q13 (4.07, green)
15b	72°	2	Green	Ladder: Pythagoras and trigonometry TY2 Q8 (2.04, green) and TY4 Q13 (4.07, green)
	Total	**33**	26 green · 7 blue	

Section 1 marking criteria – Data

Question 1 1 mark ■ Green Classifying data TY9 Q9 (9.05, green)

ANSWER **D (The time taken to swim 50 m)**
MARKS **1 mark:** D (The time taken to swim 50 m)
NOTES *Time is measured, so it is numerical continuous. A is numerical discrete (counted), B is categorical nominal, C is categorical ordinal.*

Question 2 1 mark ■ Green Shape of a distribution (dot plot) TY9 Q6a (9.03, green)

ANSWER **B (Negatively skewed, clustered at 8 and 9)**
MARKS **1 mark:** B (Negatively skewed, clustered at 8 and 9)
NOTES *Most scores are high (a cluster at 8 and 9) with a long tail to the left (low scores), so the data is negatively skewed. (Mean 7.3 is below the median 8.)*

Question 3 1 mark ■ Green Mode from a histogram TY9 Q4b (9.02, green)

ANSWER **A (2)**
MARKS **1 mark:** A (2)
NOTES *The tallest column is at 2 (frequency 9), so the mode is 2 people. 9 is the frequency, not the score.*

Question 4 1 mark ■ Green Effect of an outlier TY9 Q2c and Q3c-d (9.01, 9.02, green)

ANSWER **C (The mean)**
MARKS **1 mark:** C (The mean)
NOTES *Mean 7.9 → 6.44 (changes by 1.46); median 6.5 → 6; mode 6 → 6 (no change).*

Question 5 3 marks ■ Green Mean and median from a stem-and-leaf plot TY9 Q3b (9.02, green)

a) ANSWER **25 minutes** 2 marks · ■ Green
WORKING Sum = 12 + 15 + 18 + 21 + 24 + 24 + 27 + 30 + 33 + 35 + 36 = 275
Mean = 275 ÷ 11
MARKS **1 mark:** sum of the scores: 275
1 mark: mean 25
NOTES *Follow-through: a wrong sum divided correctly by 11 earns the second mark.*

b) ANSWER **24 minutes** 1 mark · ■ Green
WORKING 11 scores: the median is the 6th score
MARKS **1 mark:** 24

Marking criteria (continued)

Question 6 4 marks ■ Green Reading a column graph TY9 Q4 (9.02, green)

a) ANSWER **16 students** 1 mark · ■ Green
WORKING 20 − 4
MARKS **1 mark:** 16

b) ANSWER **60 students** 1 mark · ■ Green
WORKING 16 + 8 + 20 + 4 + 12
MARKS **1 mark:** 60

c) ANSWER **40%** 2 marks · ■ Green
WORKING 20 + 4 = 24 students
24 ÷ 60 × 100%
MARKS **1 mark:** 24 students chose surfing or tennis
1 mark: 40%
NOTES *Follow-through from part b.*

Question 7 6 marks ● Blue Comparing two data sets (back-to-back stem-and-leaf plot) TY9 Q7 (9.04, blue)

a) ANSWER **Kestrels: 39; Tigers: 42.5** 2 marks · ● Blue
WORKING Kestrels: (38 + 40) ÷ 2
Tigers: (41 + 44) ÷ 2
MARKS **1 mark:** Kestrels median 39
1 mark: Tigers median 42.5
NOTES *Common error: one middle score only (38 or 40; 41 or 44): 0 marks for that team.*

b) ANSWER **The Tigers: their median (42.5) is higher than the Kestrels' (39).** 1 mark · ● Blue
MARKS **1 mark:** Tigers, with a reason that compares the medians
NOTES *Follow-through from part a. Accept a comparison of where the scores cluster.*

c) ANSWER **The Kestrels: their range is 15 goals (46 – 31), much smaller than the Tigers' 35 goals (57 – 22).** 3 marks · ● Blue
WORKING Kestrels: 46 − 31 = 15
Tigers: 57 − 22 = 35
MARKS **1 mark:** Kestrels range 15
1 mark: Tigers range 35
1 mark: Kestrels, because their range is smaller
NOTES *Follow-through: the third mark for choosing the team with the smaller of the student's two ranges.*

Section 2 marking criteria – Surface area and volume

Question 1 1 mark ■ Green Metric units: capacity (kL to L) TY10 Q1i (10.01, green)

ANSWER **C (4600 L)**
MARKS **1 mark:** C (4600 L)
NOTES *1 kL = 1000 L, so 4.6 × 1000 = 4600 L. The others use × 100, × 10 000 or ÷ 1000.*

Question 2 1 mark ■ Green Precision and absolute error TY10 Q2 (10.02, green)

ANSWER **A (±25 mL)**
MARKS **1 mark:** A (±25 mL)
NOTES *The marks are 50 mL apart, so the precision is 50 mL and the absolute error is ½ × 50 = ±25 mL. B is the precision, C the gap between the numbered marks, D half of 1 mL.*

Marking criteria (continued)

Question 3 1 mark Green Area of a parallelogram TY10 Q6a (10.05, green)

ANSWER **D (72 cm²)**
MARKS **1 mark:** D (72 cm²)
NOTES *A = 12 × 6 = 72 cm² (the height is the perpendicular height, 6 cm, not the sloping side).*

Question 4 1 mark Green Area of a kite TY10 Q6b (10.05, green)

ANSWER **B (176 cm²)**
MARKS **1 mark:** B (176 cm²)
NOTES *A = ½ × 16 × 22 = 176 cm². 352 cm² forgets the ½.*

Question 5 2 marks Green Perimeter and area of a triangle TY10 Q4 and Q5 (10.04, green)

a) ANSWER **36 cm** 1 mark · Green
WORKING *P = 12 + 9 + 15*
MARKS **1 mark:** 36 cm

b) ANSWER **54 cm²** 1 mark · Green
WORKING *A = ½ × 12 × 9*
MARKS **1 mark:** 54 cm²
NOTES *Common errors: 108 cm² (no ½) or ½ × 12 × 15 = 90 cm² (uses the sloping side): 0 marks.*

Question 6 2 marks Blue Circumference and area of a circle TY10 Q7 (10.06, blue)

a) ANSWER **40.8 cm** 1 mark · Blue
WORKING *C = 2 × π × 6.5 = 40.841…*
MARKS **1 mark:** 40.8 cm

b) ANSWER **132.7 cm²** 1 mark · Blue
WORKING *A = π × 6.5² = 132.732…*
MARKS **1 mark:** 132.7 cm²
NOTES *Common error: π × 13² (uses the diameter) = 530.9 cm²: 0 marks.*

Question 7 4 marks Blue Volume and surface area of a cylinder TY10 Q12 and Q11 (10.09, 10.08, blue)

a) ANSWER **1696.5 cm³** 2 marks · Blue
WORKING *V = π × 6² × 15 = 1696.460…*
MARKS **1 mark:** correct substitution: π × 6² × 15
1 mark: 1696.5 cm³

b) ANSWER **791.7 cm²** 2 marks · Blue
WORKING *SA = 2 × π × 6² + 2 × π × 6 × 15*
= 226.19… + 565.49…
MARKS **1 mark:** correct area of the two circular ends (2 × π × 6² = 226.2 cm²) or of the curved surface (2 × π × 6 × 15 = 565.5 cm²)
1 mark: 791.7 cm²
NOTES *Leaving out one or both circular ends (678.6 or 565.5 cm²) earns the first mark only.*

Marking criteria (continued)

Question 8 5 marks ■ Green Perimeter and area of a composite shape (lawn) TY10 Q4a and Q5 (10.04, green)

a)
ANSWER **60 m** 1 mark · ■ Green
WORKING $P = 18 + 7 + 10 + 5 + 8 + 12$
MARKS **1 mark:** 60 m

b)
ANSWER **166 m²** 3 marks · ■ Green
WORKING Bottom rectangle: $18 \times 7 = 126$ m²
Top rectangle: $8 \times 5 = 40$ m²
Area = 126 + 40
MARKS **1 mark:** bottom rectangle 126 m²
1 mark: top rectangle 40 m²
1 mark: total area 166 m²
NOTES *Another split (left $8 \times 12 = 96$ m² and right $10 \times 7 = 70$ m²) or $18 \times 12 - 10 \times 5$ earns the same marks.*

c)
ANSWER **$2058.40** 1 mark · ■ Green
WORKING 166 × $12.40
MARKS **1 mark:** $2058.40
NOTES *Follow-through from part b.*

Question 9 5 marks ● Blue + ■ Green Volume, capacity and surface area of a closed box TY10 Q14 (10.09, blue), Q1i (10.01, green) and Q9a (10.07, blue)

a)
ANSWER **72 000 cm³** 1 mark · ● Blue
WORKING $V = 60 \times 40 \times 30$
MARKS **1 mark:** 72 000 cm³

b)
ANSWER **72 L** 1 mark · ■ Green
WORKING 72 000 ÷ 1000
MARKS **1 mark:** 72 L
NOTES *Follow-through from part a.*

c)
ANSWER **10 800 cm²** 3 marks · ● Blue
WORKING Top and bottom: $2 \times 60 \times 40 = 4800$ cm²
Front and back: $2 \times 60 \times 30 = 3600$ cm²
Two ends: $2 \times 40 \times 30 = 2400$ cm²
$SA = 4800 + 3600 + 2400$
MARKS **1 mark:** the three different face areas: 2400, 1800 and 1200 cm²
1 mark: each face area doubled (six faces)
1 mark: surface area 10 800 cm²
NOTES *Three faces only (5400 cm²) earns the first mark only.*

Section 3 marking criteria – Mixed Topics (Chapters 1–8)

Question 1 1 mark ■ Green Substitution TY1 Q2 (1.02, green)

ANSWER **C (47)**
MARKS **1 mark:** C (47)
NOTES *$2a^2 - b = 2 \times 5^2 - 3 = 2 \times 25 - 3 = 47$. (97 squares 2a: $(2 \times 5)^2 - 3$; 17 treats a^2 as 2a; 7 leaves out the square.)*

Marking criteria (continued)

Question 2 1 mark ■ Green Pythagoras' theorem: hypotenuse TY2 Q5 (2.03, green)

ANSWER **B (10.9)**

MARKS **1 mark:** B (10.9)

NOTES *$x^2 = 5.6^2 + 9.3^2 = 31.36 + 86.49 = 117.85$, so $x = \sqrt{117.85} = 10.856... \approx 10.9$. (14.9 adds the sides, 7.4 subtracts the squares, 3.7 subtracts the sides.)*

Question 3 1 mark ■ Green Discount and selling price TY3 Q16 (3.06, green)

ANSWER **C ($123.25)**

MARKS **1 mark:** C ($123.25)

NOTES *Discount = 15% × $145 = $21.75; selling price = $145 − $21.75 = $123.25. ($21.75 is the discount only, $130.00 takes off $15, $166.75 increases the price.)*

Question 4 1 mark ■ Green Index laws: multiplying terms TY5 Q1 (5.01, green)

ANSWER **A ($12m^8$)**

MARKS **1 mark:** A ($12m^8$)

NOTES *Multiply the numbers and add the indices: $4 \times 3 = 12$ and $m^6 \times m^2 = m^8$. ($7m^8$ adds the numbers, $12m^{12}$ multiplies the indices, $12m^4$ subtracts the indices.)*

Question 5 1 mark ■ Green Salary: pay per fortnight TY8 Q1 (8.01, green)

ANSWER **B ($3140.00)**

MARKS **1 mark:** B ($3140.00)

NOTES *Pay per fortnight = $81 640 ÷ 26 = $3140.00. ($1570.00 is the pay per week (÷ 52), $6803.33 the pay per month (÷ 12), and $3401.67 divides by 24 instead of 26.)*

Question 6 3 marks ■ Green Simplifying algebraic expressions TY1 Q3 and Q4 (1.03, 1.04, green)

a) ANSWER **$3a + 4b$** 1 mark · ■ Green

MARKS **1 mark:** $3a + 4b$

b) ANSWER **$12xy$** 1 mark · ■ Green

MARKS **1 mark:** $12xy$

c) ANSWER **$3n$** 1 mark · ■ Green

MARKS **1 mark:** $3n$

Question 7 2 marks ● Blue Expanding and simplifying TY1 Q7 (1.07, blue)

ANSWER **$10x - 3$**

WORKING $= 12x - 15 - 2x + 12$

MARKS **1 mark:** correct expansion of both brackets: $12x - 15 - 2x + 12$

1 mark: $10x - 3$

NOTES *Common error: $10x - 27$ ($-2 \times (-6)$ taken as -12): 1 mark.*

Question 8 2 marks ● Blue Factorising TY1 Q8 (1.08, blue)

a) ANSWER **$3(2x + 5)$** 1 mark · ● Blue

MARKS **1 mark:** $3(2x + 5)$

b) ANSWER **$y(y - 8)$** 1 mark · ● Blue

MARKS **1 mark:** $y(y - 8)$

Marking criteria (continued)

Question 9 2 marks ■ Green Simple interest TY3 Q19 (3.07, green)

ANSWER **$913.50**

WORKING $I = \$7250 \times 0.042 \times 3$

MARKS **1 mark:** correct substitution: $\$7250 \times 0.042 \times 3$
1 mark: \$913.50

NOTES *Common error: r = 4.2 instead of 0.042 ($91 350.00): 1 mark.*

Question 10 2 marks ■ Green Profit as a percentage of the cost price TY3 Q17 (3.06, green)

a) ANSWER **$132** 1 mark · ■ Green

WORKING $\$612 - \$480$

MARKS **1 mark:** \$132

b) ANSWER **27.5%** 1 mark · ■ Green

WORKING $\$132 \div \$480 \times 100\%$

MARKS **1 mark:** 27.5%

NOTES *Follow-through from part a. Common error: dividing by the selling price (21.6%): 0 marks.*

Question 11 5 marks ■ Green + ● Blue Solving equations TY7 Q1 (7.01, green) and Q5 (7.03, blue)

a) ANSWER **$x = 7$** 2 marks · ■ Green

WORKING $5x = 35$

MARKS **1 mark:** $5x = 35$
1 mark: $x = 7$

b) ANSWER **$m = 8$** 3 marks · ● Blue

WORKING $7m - 2m - 6 = 3m + 10$
$5m - 6 = 3m + 10$
$2m = 16$

MARKS **1 mark:** expands the bracket: $7m - 2m - 6$
1 mark: collects the terms: $2m = 16$
1 mark: $m = 8$

NOTES *Common error: -2×3 taken as $+6$, giving $m = 2$: 2 marks (follow-through).*

Question 12 2 marks ■ Green Commission TY8 Q5 (8.03, green)

ANSWER **$1690**

WORKING Commission: $2.5\% \times \$46\,800 = \$1170$
Total: $\$520 + \$1170$

MARKS **1 mark:** commission \$1170
1 mark: total pay \$1690

NOTES *Commission only ($1170) earns the first mark.*

Marking criteria (continued)

Question 13 3 marks ■ Green Wages and overtime TY8 Q2 and Q3 (8.01, 8.02, green)

a) 1 mark · ■ Green

ANSWER **$1018.40**

WORKING 38 × $26.80

MARKS **1 mark:** $1018.40

b) 2 marks · ■ Green

ANSWER **$1259.60**

WORKING Overtime: 6 × 1.5 × $26.80 = $241.20
Total: $1018.40 + $241.20

MARKS **1 mark:** overtime pay $241.20 (6 hours at time-and-a-half)
1 mark: total $1259.60

NOTES *Follow-through from part a.*
Common error: overtime at the normal rate ($1179.20 in total): 1 mark.

Question 14 3 marks ■ Green Using a formula; solving an equation TY7 Q11 (7.06, green)

a) 1 mark · ■ Green

ANSWER **$240**

WORKING $C = 60 + 15 \times 12$

MARKS **1 mark:** $240

b) 2 marks · ■ Green

ANSWER **16 weeks**

WORKING $60 + 15w = 300$
$15w = 240$
$w = 240 \div 15$

MARKS **1 mark:** equation $60 + 15w = 300$ (or $15w = 240$)
1 mark: 16 weeks

NOTES *Accept a correct answer by guess and check with working shown: 2 marks.*

Question 15 4 marks ■ Green Ladder: Pythagoras and trigonometry TY2 Q8 (2.04, green) and TY4 Q13 (4.07, green)

a) 2 marks · ■ Green

ANSWER **4.95 m**

WORKING $h^2 = 5.2^2 - 1.6^2 = 27.04 - 2.56 = 24.48$
$h = \sqrt{24.48} = 4.9477\ldots$

MARKS **1 mark:** $h^2 = 5.2^2 - 1.6^2$ (= 24.48)
1 mark: 4.95 m

NOTES *Common error: adding the squares, $\sqrt{(5.2^2 + 1.6^2)} = 5.44$: 0 marks.*

b) 2 marks · ■ Green

ANSWER **72°**

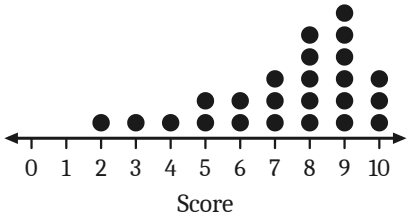
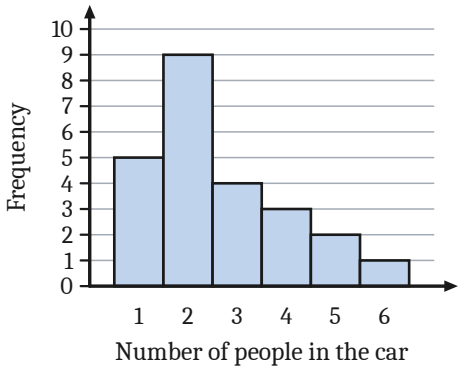
WORKING $\cos\theta = \frac{1.6}{5.2}$
$\theta = \cos^{-1}(1.6 \div 5.2) = 72.08\ldots°$

MARKS **1 mark:** $\cos\theta = 1.6 \div 5.2$ (or sin or tan with the answer to part a)
1 mark: 72°

NOTES *Accept sin or tan with the answer to part a (follow-through): both give the same angle to the nearest degree.*

Marking guide: answers in red

Section 1: Data

	Multiple Choice:	Mark
1	Which of the following is **numerical continuous** data? A. The number of goals in a hockey match B. The colours of cars in a car park C. Shirt sizes (S, M, L, XL) **D. The time taken to swim 50 m**	1
2	The dot plot shows the scores (out of 10) of 24 students in a spelling quiz. Which statement best describes the data? A. Positively skewed, clustered at 8 and 9 **B. Negatively skewed, clustered at 8 and 9** C. Negatively skewed, clustered at 2 and 3 D. Symmetrical, clustered at 8 and 9 	1
3	The histogram shows the number of people in each of 24 cars that arrived at a beach car park. What is the mode? **A. 2** B. 3.5 C. 6 D. 9 	1
4	A water polo team scored these numbers of goals in 10 games: 5, 7, 6, 8, 6, 21, 7, 5, 6, 8 Which measure of centre changes the most if the outlier, 21, is removed? A. The mode B. The median **C. The mean** D. All three change by the same amount	1

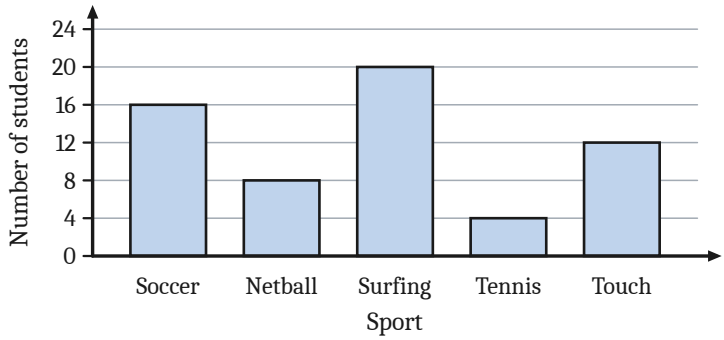
Marking guide: answers in red

Section 1: Data (continued)

Short Answer:

	Question	Working Out	Mark
5	The stem-and-leaf plot shows the number of minutes that 11 students spent on homework last night. Stem | Leaf 1 | 2 5 8 2 | 1 4 4 7 3 | 0 3 5 6 Key: 2 | 4 = 24 minutes a. Find the mean. b. Find the median. Note: Mean = sum of the scores ÷ number of scores	a. Sum = 12 + 15 + 18 + 21 + 24 + 24 + 27 + 30 + 33 + 35 + 36 = 275 Mean = 275 ÷ 11 **25 minutes** b. 11 scores: the median is the 6th score **24 minutes**	2 1

Worded Problems:

	Question	Working Out	Mark
6	Year 9 students each chose one sport for Friday afternoon. The column graph shows how many students chose each sport.  		
a	How many more students chose surfing than tennis?	20 – 4 **16 students**	1
b	How many students were surveyed altogether?	16 + 8 + 20 + 4 + 12 **60 students**	1
c	What percentage of the students chose surfing or tennis?	20 + 4 = 24 students 24 ÷ 60 × 100% **40%**	2

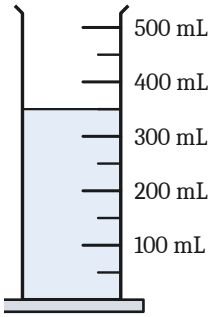
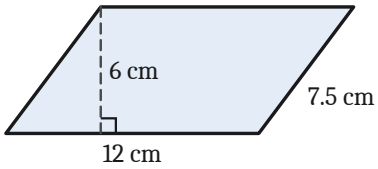
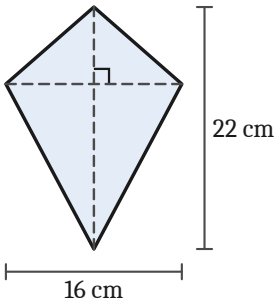
Marking guide: answers in red

Section 1: Data (continued)

	Question	Working Out	Mark
7	The Kestrels and the Tigers are two netball teams. The back-to-back stem-and-leaf plot shows the number of goals each team scored in its 12 games this season. Note: Range = highest score − lowest score	Kestrels | Stem | Tigers 2 | 2 7 8 8 7 5 4 1 | 3 | 3 6 9 6 4 3 2 1 0 | 4 | 1 4 5 7 5 | 0 3 7 Key: 4 | 1 = 41 goals	
a	Find the median number of goals for each team.	Kestrels: (38 + 40) ÷ 2 Tigers: (41 + 44) ÷ 2 **Kestrels: 39; Tigers: 42.5**	2
b	Which team usually scores more goals? Give a reason.	**The Tigers: their median (42.5) is higher than the Kestrels' (39).**	1
c	Which team is more consistent? Find the range of each team to explain.	Kestrels: 46 − 31 = 15 Tigers: 57 − 22 = 35 **The Kestrels: their range is 15 goals (46 − 31), much smaller than the Tigers' 35 goals (57 − 22).**	3
	Total		**/17**

Marking guide: answers in red

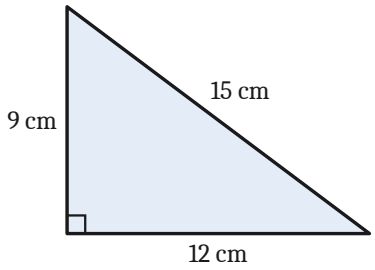
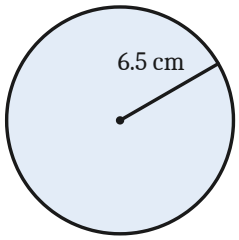
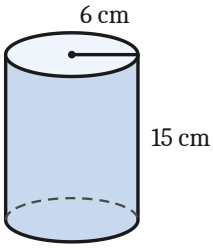
Section 2: Surface area and volume

	Multiple Choice:	Mark
1	A rainwater tank holds 4.6 kL of water. How many litres is this? A. 460 L B. 46 000 L **C. 4600 L** D. 0.0046 L	1
2	The scale on this measuring cylinder is shown. What is the absolute error of a reading taken with this measuring cylinder? Note: absolute error = ½ × precision (the smallest marked unit on the scale) **A. ±25 mL** B. ±50 mL C. ±100 mL D. ±0.5 mL 	1
3	What is the area of this parallelogram? Note: Area of a parallelogram: $A = bh$ A. 90 cm^2 B. 36 cm^2 C. 39 cm^2 **D. 72 cm^2** 	1
4	What is the area of this kite? Note: Area of a kite: $A = \frac{1}{2}xy$ (x and y are the diagonals) A. 352 cm^2 **B. 176 cm^2** C. 88 cm^2 D. 38 cm^2 	1

Marking guide: answers in red

Section 2: Surface area and volume (continued)

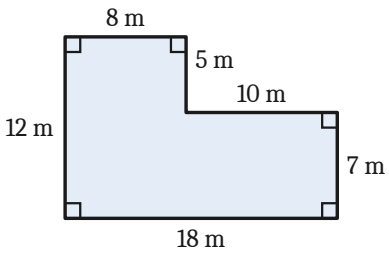
Short Answer:

	Question	Working Out	Mark
5	For this right-angled triangle, find:  a. its perimeter b. its area. Note: Area of a triangle: $A = \frac{1}{2}bh$	a. $P = 12 + 9 + 15$ **36 cm** b. $A = \frac{1}{2} \times 12 \times 9$ **54 cm²**	1 1
6	For this circle, find, correct to 1 decimal place:  a. its circumference b. its area. Note: Circumference: $C = 2\pi r$ Note: Area: $A = \pi r^2$	a. $C = 2 \times \pi \times 6.5 = 40.841\ldots$ **40.8 cm** b. $A = \pi \times 6.5^2 = 132.732\ldots$ **132.7 cm²**	1 1
7	This closed cylinder has a radius of 6 cm and a height of 15 cm.  a. Find its volume, correct to 1 decimal place. b. Find its surface area, correct to 1 decimal place. Note: Volume: $V = \pi r^2 h$ Note: Surface area: $SA = 2\pi r^2 + 2\pi rh$	a. $V = \pi \times 6^2 \times 15 = 1696.460\ldots$ **1696.5 cm³** b. $SA = 2 \times \pi \times 6^2 + 2 \times \pi \times 6 \times 15$ $= 226.19\ldots + 565.49\ldots$ **791.7 cm²**	2 2

Marking guide: answers in red

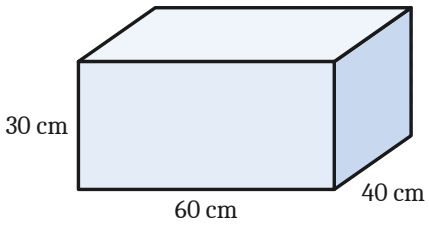
Section 2: Surface area and volume (continued)

Worded Problems:

	Question	Working Out	Mark
8	The plan shows an L-shaped lawn at a primary school. All of the corners are right angles. Note: Area of a rectangle: $A = lb$		
a	Garden edging is laid around the outside of the lawn. How many metres of edging are needed?	$P = 18 + 7 + 10 + 5 + 8 + 12$ **60 m**	1
b	Find the area of the lawn.	Bottom rectangle: $18 \times 7 = 126$ m^2 Top rectangle: $8 \times 5 = 40$ m^2 Area $= 126 + 40$ **166 m^2**	3
c	New turf costs \$12.40 per square metre. Find the cost of the turf for the whole lawn.	$166 \times \$12.40$ **\$2058.40**	1

Marking guide: answers in red

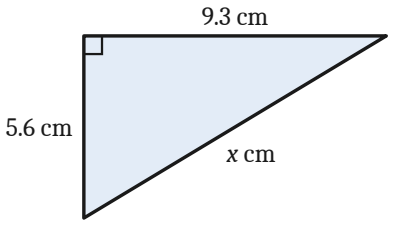
Section 2: Surface area and volume (continued)

	Question	Working Out	Mark
9	A closed wooden storage box is a rectangular prism 60 cm long, 40 cm wide and 30 cm high. 		
a	Find the volume of the box in cm^3. Note: Volume of a rectangular prism: $V = lwh$	$V = 60 \times 40 \times 30$ **$72\,000\ cm^3$**	1
b	How many litres does the box hold? Note: 1 L = 1000 cm^3	$72\,000 \div 1000$ **72 L**	1
c	The outside of the box is to be painted. Find the surface area of the box. Note: Surface area = the total area of the 6 faces; area of a rectangle: $A = lb$	Top and bottom: $2 \times 60 \times 40 = 4800\ cm^2$ Front and back: $2 \times 60 \times 30 = 3600\ cm^2$ Two ends: $2 \times 40 \times 30 = 2400\ cm^2$ $SA = 4800 + 3600 + 2400$ **$10\,800\ cm^2$**	3
	Total		**/22**

Marking guide: answers in red

Section 3: Mixed Topics (Chapters 1–8)

Multiple Choice:

		Mark
1	If $a = 5$ and $b = 3$, evaluate $2a^2 - b$. A. 7 B. 17 **C. 47** D. 97	1
2	Find the value of x. Select the closest answer. Note: Pythagoras' theorem: $c^2 = a^2 + b^2$ A. 14.9 **B. 10.9** C. 7.4 D. 3.7 	1
3	A skateboard is marked at \$145. In a sale it is discounted by 15%. What is the selling price? A. \$166.75 B. \$130.00 **C. \$123.25** D. \$21.75	1
4	Simplify $4m^6 \times 3m^2$. **A. $12m^8$** B. $7m^8$ C. $12m^{12}$ D. $12m^4$	1
5	Ella is a nurse. She earns an annual salary of \$81 640. How much is she paid per fortnight? Note: 1 year = 52 weeks = 26 fortnights = 12 months A. \$1570.00 **B. \$3140.00** C. \$3401.67 D. \$6803.33	1

Short Answer:

	Question	Working Out	Mark
6	Simplify: a. $5a + 3b - 2a + b$ b. $4x \times 3y$ c. $18mn \div 6m$	a. **$3a + 4b$** b. **$12xy$** c. **$3n$**	1 1 1

Marking guide: answers in red

	Section 3: Mixed Topics (Chapters 1–8) (continued)		
	Question	**Working Out**	**Mark**
7	Expand and simplify: $3(4x - 5) - 2(x - 6)$	$= 12x - 15 - 2x + 12$ **$10x - 3$**	2
8	Factorise: a. $6x + 15$ b. $y^2 - 8y$	a. **$3(2x + 5)$** b. **$y(y - 8)$**	1 1
9	Liam invests \$7250 for 3 years at 4.2% p.a. simple interest. Calculate the interest he earns. Note: Simple interest: $I = Prn$	$I = \$7250 \times 0.042 \times 3$ **\$913.50**	2
10	A surf shop buys a surfboard for \$480 and sells it for \$612. a. Find the profit. b. Find the profit as a percentage of the cost price. Note: Percentage profit = profit ÷ cost price × 100%	a. \$612 − \$480 **\$132** b. \$132 ÷ \$480 × 100% **27.5%**	1 1

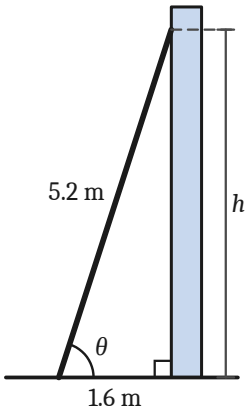
Marking guide: answers in red

Section 3: Mixed Topics (Chapters 1–8) (continued)

	Question	Working Out	Mark
11	Solve each equation. a. $5x - 8 = 27$ b. $7m - 2(m + 3) = 3m + 10$	a. $5x = 35$ **$x = 7$** b. $7m - 2m - 6 = 3m + 10$ $5m - 6 = 3m + 10$ $2m = 16$ **$m = 8$**	2 3
12	Jake sells cars. He is paid $520 per week plus 2.5% commission on his sales. Last week he sold $46 800 worth of cars. Calculate Jake's total pay for the week.	Commission: 2.5% × $46 800 = $1170 Total: $520 + $1170 **$1690**	2
	Worded Problems:		
	Question	**Working Out**	**Mark**
13	Mia works in a bakery. Her normal rate of pay is $26.80 per hour for the first 38 hours each week. She is paid time-and-a-half for each hour after that. Last week Mia worked 44 hours.		
a	Calculate Mia's pay for the first 38 hours.	38 × $26.80 **$1018.40**	1
b	Calculate Mia's total pay for the 44 hours.	Overtime: 6 × 1.5 × $26.80 = $241.20 Total: $1018.40 + $241.20 **$1259.60**	2

Marking guide: answers in red

Section 3: Mixed Topics (Chapters 1–8) (continued)

	Question	Working Out	Mark
14	A gym charges a joining fee of \$60 plus \$15 per week. The total cost, C dollars, for w weeks is given by the formula **$C = 60 + 15w$**		
a	Find the total cost for 12 weeks.	$C = 60 + 15 \times 12$ **\$240**	1
b	Jo paid \$300 altogether. Write an equation and solve it to find how many weeks Jo went to the gym.	$60 + 15w = 300$ $15w = 240$ $w = 240 \div 15$ **16 weeks**	2
15	A ladder 5.2 m long leans against a vertical wall. The foot of the ladder is 1.6 m from the wall, on level ground. Note: Pythagoras' theorem: $c^2 = a^2 + b^2$ Note: $\sin\theta = \frac{\text{opp}}{\text{hyp}}$, $\cos\theta = \frac{\text{adj}}{\text{hyp}}$, $\tan\theta = \frac{\text{opp}}{\text{adj}}$ 		
a	Find h, the height the ladder reaches up the wall, correct to 2 decimal places.	$h^2 = 5.2^2 - 1.6^2 = 27.04 - 2.56 = 24.48$ $h = \sqrt{24.48} = 4.9477\ldots$ **4.95 m**	2
b	Find the angle θ that the ladder makes with the ground, correct to the nearest degree.	$\cos\theta = \frac{1.6}{5.2}$ $\theta = \cos^{-1}(1.6 \div 5.2) = 72.08\ldots°$ **72°**	2
	Total		**/33**